

SubGHz-Bridge

How it works — technical mini-manual

IP link over a 915 MHz GFSK radio · Elecrow ThinkNode M5 (ESP32-S3 + SX1262)

It is a **GFSK** sub-GHz link at 915 MHz, built around the **SX1262** transceiver, that carries **Ethernet frames**: a custom **MAC**, half-duplex via **TDD**, with a stop-and-wait **ARQ**, fragments each frame —because an Ethernet frame of up to 1514 bytes does not fit in a 246-byte radio packet— and acknowledges every fragment over the air; on top of that, the IP stack performs **double NAPT** to reach the internet. In practice, you browse at the speed of a 1990s dial-up modem (≈ 55 kbps).

1 · What it is and what it is for

Two identical nodes —same hardware, role chosen in software— that bring IP connectivity to a place with no WiFi coverage, using low-frequency radio that propagates much farther. One node (the **client**) creates a WiFi network for a phone or PC; the other (the **gateway**) connects to the home router. Between them, where cable or WiFi would normally sit, there is a 915 MHz radio link. It is enough for messaging, webmail and light browsing; not for streaming.

2 · How it works, layer by layer

2.1 Physical layer (PHY)

Everything that is the actual radio signal. It uses GFSK modulation in the **sub-GHz** band (below 1 GHz), in the license-free **ISM** band at 915 MHz. Compared with WiFi's 2.4 GHz, sub-GHz propagates better and penetrates more obstacles at the same power; the trade-off is less bandwidth. The transceiver is the Semtech **SX1262**. Parameters: 250 kbps, frequency deviation 80 kHz, Gaussian filter BT 0.5, receive bandwidth 467 kHz, +20 dBm output, TCXO (temperature-compensated crystal oscillator) clock.

2.2 Data-link layer (MAC)

It decides who transmits and when, builds the frames and handles the acknowledgements. Since there is a single channel and a single transceiver, the link is **half-duplex** (it transmits or receives, never both at once) using **TDD** the two nodes take turns in time, like a walkie-talkie. The **gateway** acts as master (it transmits and then listens) and the **client** as slave (it listens and replies within the same cycle).

Reliability comes from a stop-and-wait **ARQ** each fragment is confirmed with an **ACK** and retransmitted if lost (up to 3 times). The ACK travels **piggyback**, i.e. inside the header of the return fragment, without spending a separate transmission. “Stop-and-wait” means a single fragment in flight at a time: simple and robust, but it pays a full round trip per fragment.

2.3 Fragmentation and Ethernet-frame transport

An Ethernet frame can be up to 1514 bytes —an **MTU** of 1500 data bytes plus 14 of Ethernet header— and only 246 bytes fit in a radio packet. That is why the MAC **fragments**: it splits the frame into 240-byte data chunks, each with its own 6-byte header, and reassembles them on the far side. The control bits **FIRST** (first fragment) and **MF (More Fragments)** mark the start and the continuation; since stop-and-wait guarantees ordering, there is no need to number each fragment.

Key point: what gets fragmented is the whole **Ethernet frame (layer 2)**, not the IP packet. The system **encapsulates (tunnels)** Ethernet over the radio link. This is **link-layer fragmentation**, a mechanism of its own,

distinct from **IP fragmentation** (the Identification/Offset fields of the IP header, which are left untouched here). The IP packet rides intact inside and never knows it crossed a radio.

2.4 Network layer (IP)

The radio presents itself to the network stack as just another Ethernet interface. Because it carries complete Ethernet frames, routing, addressing and network services work without modifying the stack. The client runs a **DHCP** server that hands the phone its IP, mask, gateway and DNS automatically. **DNS** translates names (for example google.com) into IP addresses.

Internet access is achieved with **NAT** —translating private addresses to another network when crossing from one to another— and, more precisely, **NAPT** the “NAT with ports” that any home router uses: it multiplexes many private addresses behind a single one, telling them apart by port number. Here there are **two NAPT stages in cascade**: the client masks the phone's network (192.168.5.x) behind its own radio-link address (10.0.0.2), and the gateway masks it again behind the address the home router gave it (192.168.1.x). The phone ends up browsing as just another device on the home network.

3 · What exactly gets fragmented

- The **Ethernet frame (layer 2)** is fragmented, not the IP packet (layer 3).
- It is fragmented because the frame is **large** (up to 1514 B) and the radio packet is **small** (246 B).
- An ACK confirms **each fragment**.
- It is **link-layer** fragmentation.

4 · Performance and why

Measured metrics: usable IP throughput ≈ 55 kbps; latency to the internet **RTT** ≈ 256 ms; verified with a ping to 8.8.8.8 and with Telegram working. The PHY delivers 250 kbps raw; the gap down to ≈ 55 kbps usable is the overhead of half-duplex plus stop-and-wait: the round trip per fragment, the TX↔RX switching, and the clock start-up on every transition. The next optimization step is a **windowed ARQ** (several fragments in flight before the ACK).

5 · Reference parameters

Band	915 MHz — ISM sub-GHz (license-free)
Modulation	GFSK, 250 kbps
Deviation (fdev)	80 kHz · Gaussian filter BT 0.5
RX bandwidth	467 kHz
TX power	+20 dBm
Transceiver	Semtech SX1262
Radio frame	246 B (6 header + 240 data)
Time-on-Air / frame	≈ 8.22 ms
Duplexing	TDD half-duplex, master–slave
Reliability	stop-and-wait ARQ · piggyback ACK · 3 retries
Transport	Encapsulated Ethernet frames · double NAPT
Usable IP throughput	≈ 55 kbps
Latency (WAN RTT)	≈ 256 ms
Platform	Elecrow ThinkNode M5 (ESP32-S3 + SX1262)